

Atmospheric oxygen constraints on Southern Ocean productivity and drivers of carbon uptake

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Ocean net primary production fixes dissolved carbon into organic matter while producing O₂, driving the biological carbon pump that contributes to ocean CO₂ uptake. The Southern Ocean plays a critical role in carbon export, yet its productivity estimates remain highly uncertain due to limited observations. Here we constrain Southern Ocean (south of -44° S) net primary production by linking Coupled Model Intercomparison Project Phase 6 (CMIP6)-modelled productivity to modelled air–sea O₂ fluxes and applying O₂ flux estimates derived from airborne O₂/N₂ observations. We find an annual net primary production of 6.5 ± 1.36 PgC yr⁻¹, substantially higher than most CMIP6 model and satellite-based estimates, but consistent with Argo oxygen-based estimates. We show that CMIP6 models with underestimated productivity exhibit weak summer CO₂ uptake, with some also showing excessive summer temperature-driven outgassing. Together, these models produce incorrect seasonal CO₂ flux cycles with summer outgassing, whereas observation-based estimates indicate summer uptake. These errors may stem from inadequate model representation of ocean vertical mixing, which affects nutrient supply, stratification and heat redistribution. Our productivity estimates provide quantitative benchmarks that, combined with constraints from airborne CO₂ observations and surface ocean pCO₂ and temperature observations, reduce uncertainty in estimates of model-projected end-of-century Southern Ocean CO₂ uptake by 53%.

Ocean net primary production (NPP) represents dissolved inorganic carbon (DIC) fixed by photosynthetic organisms minus the carbon lost through their respiration¹. NPP is fundamentally linked to photosynthesis, which simultaneously produces O₂ in a nearly fixed stoichiometric ratio with carbon fixation. Concurrently, microbial and zooplankton respiration oxidizes 50–90% of the fixed organic carbon back to DIC while consuming oxygen, leaving net community production (NCP) at 10–50% of NPP^{2–4}. The remaining organic carbon sinks into the deep ocean via the ‘biological carbon pump’, sequestering carbon away from the atmosphere^{2,5}.

The Southern Ocean (SO) is important for determining global oceanic carbon uptake, nutrient supply and productivity, but remains undersampled⁶. Coupled Model Intercomparison Project Phase 6 (CMIP6) models⁷ diverge substantially in estimates of SO productivity, carbon pump strength, and seasonal and annual CO₂ flux, with errors often correlated^{8–10}. While NCP is more directly linked to biological CO₂ uptake and net O₂ production, most CMIP6 models did not report NCP. Instead, we focus on NPP, which quantifies photosynthetic production with broader implications, but we also present a limited NCP analysis for models reporting relevant variables.

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Recent observation-based SO NPP estimates are derived from diel cycles of oxygen measured by Argo floats¹¹. Satellites support SO NPP estimates with broad coverage based on chlorophyll-*a* (Chl-*a*) concentrations estimated from ocean colour measurements^{12,13}. However, satellite-based approaches may underestimate NPP owing to cloud cover and the inability to detect subsurface Chl-*a* maxima that contribute substantially to biological activity^{14–16}. These products diverge substantially, with Argo floats yielding about 6.3 PgC yr⁻¹ while satellite-based estimates range from 2.5 to 4.5 PgC yr⁻¹ (refs. 11–13,17–20).

Air–sea O₂ flux can serve as a proxy for NPP due to the tight link through photosynthesis, as revealed by biogeochemical Argo data²¹. Net O₂ flux is driven by thermal and non-thermal processes that reinforce the flux seasonal cycle. During spring to summer, surface warming reduces O₂ solubility, while NPP leads to biological O₂ production; together, these processes create O₂ supersaturation and promote outgassing. NPP is therefore expected to correlate well with the non-thermal component of O₂ efflux, even though some biologically produced O₂ is consumed through respiration.

Past studies have estimated the seasonal cycle of air–sea O₂ flux across the SO from surface ocean measurements using extrapolation or averaging approaches^{21–23}, but might be limited by gas exchange parameterization and sparse sampling²⁴. The SO seasonal air–sea O₂ flux has also been resolved as a combined O₂ and CO₂ flux from atmospheric measurements through Bayesian inverse modelling²⁵. However, this approach brings uncertainty related to the use of only three flask stations spanning from the South Pole to 40° S (Extended Data Fig. 1), atmospheric transport model biases^{26,27}, and requirements to subtract prior CO₂ flux estimates.

Airborne observations enable precise estimation of large-scale flux features by capturing basin-wide signals through rapid atmospheric mixing while overcoming transport biases that limit the interpretation of surface measurements. An isentrope-aligned box model was developed to quantify tracer surface fluxes using airborne measurements^{26,28,29}, with atmospheric mixing diagnosed using energy budgets from reanalysis. The method effectively resolves seasonal CO₂ fluxes over three SO latitude bands²⁶, using airborne CO₂ data over the Pacific and Atlantic^{30–32}. Here, we apply this method to provide robust estimates of seasonal air–sea O₂ fluxes over the SO (south of -44° S), using δ(O₂/N₂) measurements from the same airborne campaigns. The resolved seasonal O₂ flux estimates constrain SO NPP in CMIP6 models through a hierarchical emergent constraint (HEC³³). This constraint reveals that most CMIP6 models substantially underestimate annual SO NPP, leading to weak summer CO₂ uptake. Many models also overestimate thermal-driven CO₂ outgassing, further biasing the seasonal cycle. Our results demonstrate how atmospheric O₂ observations can diagnose biological productivity in Earth system models, providing a pathway for improving model projections of ocean carbon dynamics.

Seasonal air–sea O₂ flux inferred from airborne observations

Previous aircraft projects, including the HIPPO³⁰, 2009–2011 project, the O₂/N₂ Ratio and CO₂ Airborne Southern Ocean (ORCAS³¹, 2016) study and the Atmospheric Tomography (ATom³², 2016–2018) mission, provide meridional and vertical profiles of δ(O₂/N₂)*, which is a scaled sum of atmospheric O₂, N₂ and Ar concentrations (Methods) primarily reflecting variations in atmospheric O₂ concentration. We observe strong δ(O₂/N₂)* gradients across moist isentropes (θ_e) above the SO during both summer and winter, with the gradients reversing sign seasonally (Fig. 1a,b and Extended Data Fig. 2), indicating clear sources and sinks of ocean O₂ moderated by atmospheric mixing.

Here, we provide a SO air–sea O₂ flux seasonal climatology (2009–2018 average) derived using atmospheric δ(O₂/N₂)* data from 12 airborne campaigns. We first quantify δ(O₂/N₂)* flux, which

reflects the combined influence of surface O₂, N₂ and Ar fluxes on atmospheric δ(O₂/N₂)*. We then isolate the oceanic O₂ flux by correcting for the small impact of air–sea N₂ and Ar fluxes, and adjusting for minor non-oceanic O₂ fluxes. Surface δ(O₂/N₂)* flux is resolved by a mass-indexed θ_e-aligned (*M*_{θ_e}-aligned) box model that separates the troposphere into boxes bounded by *M*_{θ_e} surfaces (Methods). We use a single box for flux calculation from the South Pole to -44° S as defined by *M*_{θ_e} < 30 × 10¹⁶ kg, and a box further to the north as the boundary condition (Supplementary Fig. 1). The method calculates flux by balancing observed atmospheric δ(O₂/N₂)* inventory changes against surface flux plus cross-*M*_{θ_e} δ(O₂/N₂)* transport via diabatic mixing, with the mixing rate diagnosed from the moist static energy budgets of four reanalyses. Our air–sea O₂ flux estimates account for uncertainty in δ(O₂/N₂)* measurements, in methodological adjustments made to the airborne data, in small air–sea N₂, Ar and non-oceanic O₂ flux corrections (Extended Data Fig. 3), in a flux temporal bias correction due to incomplete coverage from 2009 to 2018, and in a spatial bias correction applied to the airborne data.

We show net air–sea O₂ fluxes calculated from each airborne campaign, projected to a mean seasonal cycle, in Fig. 1c (black points). The fitted flux seasonal cycle has peak outgassing in January with a net outgassing period extending from October through May. This pattern results from thermal-induced solubility changes (Fig. 1c, red) and non-thermal (Fig. 1c, blue) processes that act in phase to strengthen the net O₂ flux cycle. We quantify the thermal flux component by scaling ocean heat flux, with an additional correction for incomplete equilibration and mixing-induced supersaturation due to nonlinear solubility–temperature dependence and subsurface solar heating (Methods). The residual non-thermal flux, reflecting biological processes, extends net outgassing by 3 weeks into late summer. This extended non-thermal outgassing probably reflects late-summer entrainment of O₂-rich sub-mixed layer water, where summer stratification traps photosynthetically produced O₂ below the mixed layer until late-season deepening enables outgassing^{21,34,35}. In winter, thermal and non-thermal O₂ uptake are highly correlated, as decreasing sea surface temperature (SST) and entrainment of O₂-depleted deep water are both controlled by radiative cooling²².

Constraining SO air–sea O₂ flux and NPP in CMIP6 models

We first explore the representation of air–sea O₂ flux using two sets of historical CMIP6 Earth system model simulations: (1) coupled simulations with prescribed historical forcing and atmospheric CO₂, and (2) hindcast simulations from the Ocean Model Intercomparison Project (OMIP), which are ocean-only simulations forced by a reanalysis of atmospheric observations (Methods). Output from the CMIP6 simulations reveals model structural differences in the seasonal cycle of net O₂ fluxes (Fig. 2a), with the three Max Planck Institute for Meteorology (MPI)-ESM models consistently producing the strongest and earliest onset of summer O₂ outgassing, while other models have seasonal O₂ flux phasing that aligns with our airborne-based estimate but show smaller seasonal amplitudes and shorter outgassing periods. The cross-model spreads and model-observation biases stem mainly from non-thermal processes (Fig. 2d), as thermal fluxes (Fig. 2c) are consistent across models and generally agree with reanalysis-based estimates in phase. Nevertheless, the peak summer thermal O₂ fluxes are underestimated in models by 4% to 28%.

Models with larger seasonal amplitudes of non-thermal O₂ flux also exhibit larger depth-integrated NPP (Fig. 2b). To relate O₂ flux to annual NPP over the SO, we calculate the seasonal net outgassing (SNO) of the non-thermal O₂ flux cycle (SNO_{bio}) during the growing season³⁶, a metric predominantly controlled by growing-season biological activities³⁵. SNO_{bio} is calculated as the area under the positive portion of the 2-harmonic seasonal cycle fitted to the non-thermal O₂ fluxes with the annual mean of the fit removed²³. In Fig. 3, we show that

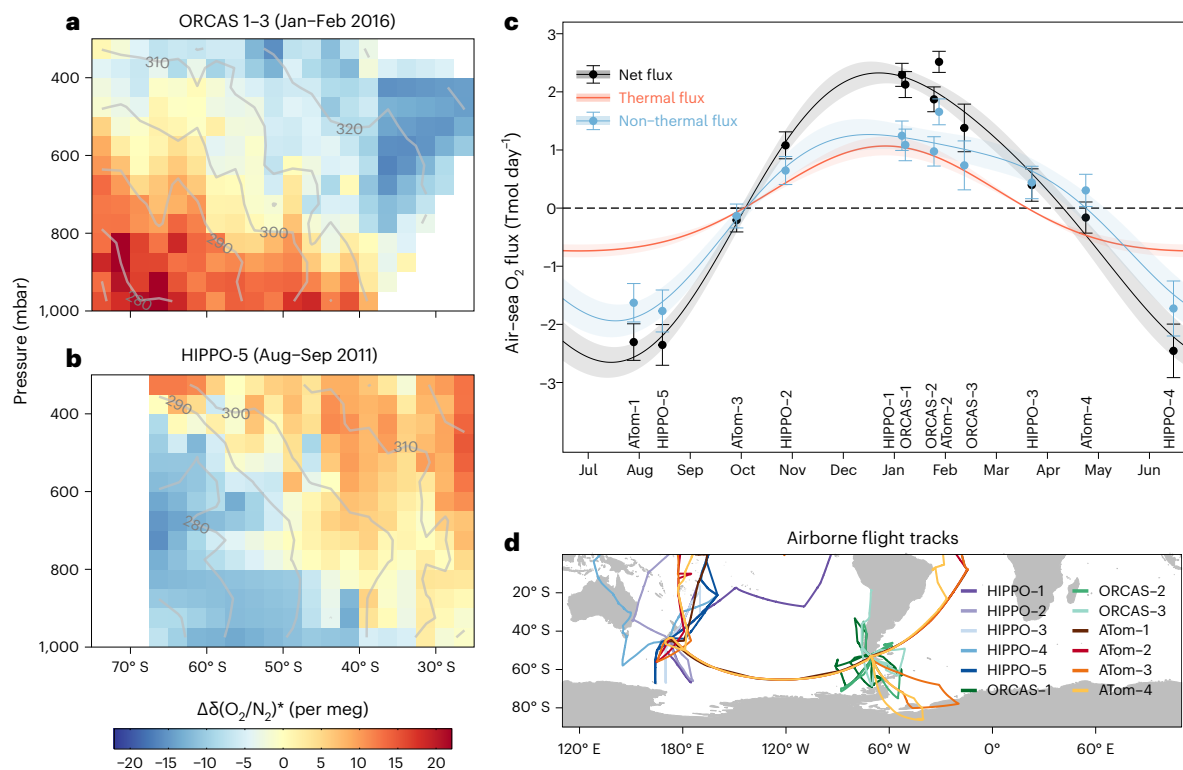


Fig. 1 | Airborne observations of $\delta(\text{O}_2/\text{N}_2)^*$ latitude–pressure patterns above the SO and airborne-resolved seasonal air–sea O_2 fluxes south of $\sim 44^\circ \text{S}$. **a, b,** Cross-sections of airborne $\delta(\text{O}_2/\text{N}_2)^*$ observations during ORCAS (summer 2016) (**a**) and HIPPO-5 (winter 2011) (**b**). Colours show the observed $\delta(\text{O}_2/\text{N}_2)^*$ grouped into 2.5° latitude and 50-mbar pressure boxes with campaign averages ($75\text{--}25^\circ \text{S}$, 1,000–300 mbar) removed. For boxes with fewer than seven observations (Supplementary Fig. 2), we used the Akima function⁴⁹ to extrapolate values solely for visualization purposes in the figure. Grey contour lines show the observed θ_e . The latitude–pressure $\delta(\text{O}_2/\text{N}_2)^*$ patterns of all campaigns are shown in Extended Data Fig. 2. **c,** Seasonal cycle of air–sea O_2 fluxes resolved from airborne observations, and the thermal and non-thermal components. Black points show ensemble mean net air–sea O_2 fluxes calculated from $\delta(\text{O}_2/\text{N}_2)^*$ measurements across 12 airborne campaigns (labelled on the x axis), with error bars indicating

standard deviation from the resolved flux ensemble ($n = 1,000$). The black line shows the mean of weighted 2-harmonic fits across ensemble members, with grey shading indicating standard deviation. Red and blue lines and shading follow the same convention for thermal and non-thermal components, respectively. Blue points show residual ensemble mean non-thermal flux estimates for each campaign, with error bars representing standard deviation. Positive flux values indicate O_2 outgassing to the atmosphere. **d,** Southern Hemisphere horizontal flight tracks from HIPPO (2009–2011), ORCAS (2016) and ATom (2016–2018) campaigns. Details of spatial and temporal coverage for each campaign are summarized in Supplementary Table 1. Basemap data in **d** from Thematic Mapping World Borders (TM_WORLD_BORDERS_v0.3; https://www.star.nesdis.noaa.gov/data/smcd1/vhp/GIS/TM_World_Borders/).

NPP linearly correlates ($r = 0.86$, $P < 0.01$) with SNO_{bio} across CMIP6 models. This correlation remains consistent regardless of NPP integration depth or timeframe (that is, annual versus summer), as most of the production occurs in the euphotic zone during the growing season (Supplementary Figs. 3 and 4).

We apply an HEC³³ framework to constrain NPP using the observed SNO_{bio} values of $186 \pm 25.2 \text{ Tmol yr}^{-1}$. HEC is based on a Bayesian statistical model that integrates both model and observational uncertainties through conditional-probability distributions³³ (Methods). We report a constrained annual NPP (south of $\sim 44^\circ \text{S}$) of $6.5 \pm 1.36 \text{ PgC yr}^{-1}$, which reduces 66% of the uncertainty in the CMIP6 model simulations (equation (13)). Among the 22 simulations analysed, 17 have annual NPP values that fall below our constrained NPP range, while the 3 MPI-ESM simulations and the UKESM1-0-LL simulation exceed this range. Our constrained NPP aligns with that derived from Argo oxygen measurements¹¹, which average 6.3 PgC yr^{-1} from 40°S to 60°S , despite slightly different latitudinal coverage. However, our estimate is significantly larger than those derived from satellite observations of Chl-*a* over the SO south of 40°S , which range from 2.5 to 4.5 PgC yr^{-1} (refs. 11–13,17–20).

We find that ACCESS-ESM1-5 and UKESM1-0-LL notably diverge from the NPP– SNO_{bio} relationship. However, exploring their mechanisms is beyond the scope of this study. Excluding these two models

from HEC would not affect our central estimate but would reduce the uncertainty to $\pm 0.96 \text{ PgC yr}^{-1}$. We acknowledge potential systematic bias in the NPP– SNO_{bio} relationship, arising from shared model structural deficiencies, common parameter choices or forcings³⁷. However, the analysed models have diverse representations of ocean biogeochemical processes, as reflected in the model e-ratio—calculated as the export production divided by NPP—which ranges from 0.11 to 0.43 (Supplementary Fig. 6). This wide range suggests we capture significant structural variability across models, and therefore, we do not incorporate additional structural uncertainty in our reported NPP constraints.

We expect that SNO_{bio} is more directly linked to net O_2 production after accounting for respiratory losses (that is, NCP). NCP can be estimated directly from the model variable ‘fbddtdic’, which represents the rate of DIC change due to biological activity, integrated over the top 100 m. However, only eight models and experiments analysed here report this variable, and its fixed 100-m integration may not align with the actual mixed layer depth³⁸. Nevertheless, we find a high correlation ($r = 0.97$) between SNO_{bio} and growing season (October to March) NCP derived from the variable ‘fbddtdic’ across these eight models (Extended Data Fig. 4), further supporting the mechanistic link between atmospheric O_2 signals and net biological production. Using the observed SNO_{bio} of $186 \pm 25.2 \text{ Tmol yr}^{-1}$, the constrained growing-season NCP amounts to $1.5 \pm 0.2 \text{ PgC yr}^{-1}$. Among these eight

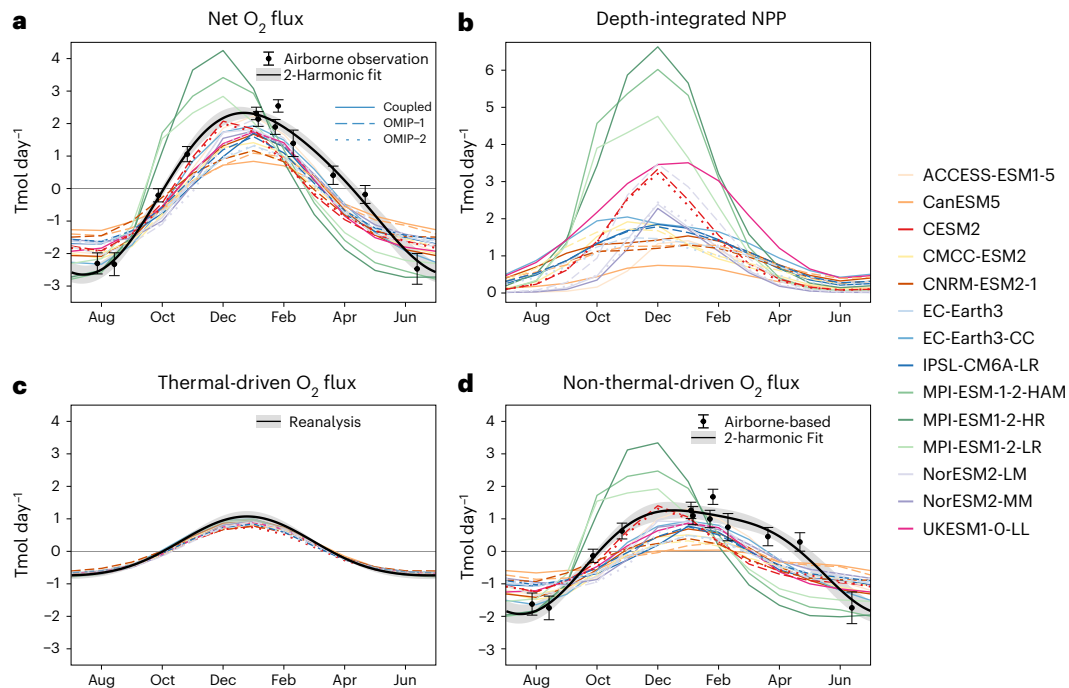


Fig. 2 | Observations and CMIP6 model simulations of seasonal air-sea O_2 fluxes and NPP south of 44° S. **a, Model net air-sea O_2 flux compared with airborne-based estimates and 2-harmonic fits (black). **b**, Model depth-integrated NPP. **c**, Thermal-driven O_2 flux from models and from reanalysis-based estimates (black), all derived from ocean heat fluxes. **d**, Model residual non-thermal O_2 flux compared with airborne-based estimates and 2-harmonic fits (black). Positive flux values indicate O_2 outgassing into the atmosphere. Black points and error bars represent the mean and standard deviation of corresponding observation-based flux estimates, while lines and grey shadings represent the 2-harmonic**

fitted seasonal flux cycle, following the same convention as in Fig. 1c. The black line and grey shading in **c** represent the mean and standard deviation computed across four reanalyses. CMIP6 simulations include historical coupled simulations (solid lines), OMIP-1 (dashed lines) and OMIP-2 (dotted lines), with each model simulation represented by a distinct colour (see details in Supplementary Table 2). Model seasonal cycles are monthly averages from 2005–2014 (coupled and OMIP-2) or 2005–2009 (OMIP-1), while our airborne data were collected between 2009 and 2018. CMIP6 data used are available through the Earth System Grid Federation (ESGF; <https://esgf-node.llnl.gov/projects/esgf-llnl/>).

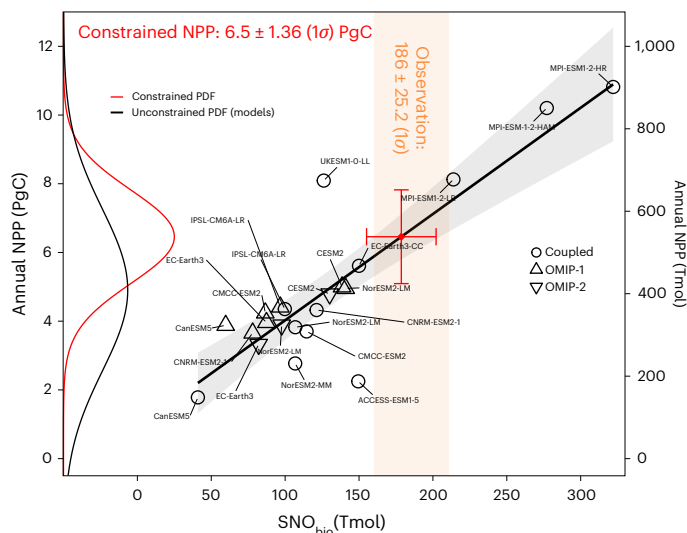


Fig. 3 | Emergent constraint on annual SO NPP using the SNO of the non-thermal O_2 flux cycle (SNO_{bio}). The scatter plot illustrates the relationship between SNO_{bio} (x axis) and annual depth-integrated NPP (y axis) for CMIP6 models and experiments. Coupled runs are shown as circles, OMIP-1 as upward triangles and OMIP-2 as downward triangles. The black line represents the linear regression, with grey shading indicating the standard error of the fit. The vertical orange shading indicates airborne-based SNO_{bio} mean and standard deviation (186 ± 25.2 Tmol), calculated from Fig. 1c. The red point with error bars denotes the mean and standard deviation of constrained NPP (6.5 ± 1.36 PgC yr $^{-1}$) and SNO_{bio} derived using HEC (Methods). Probability density functions (PDFs) for constrained (red) and unconstrained (black, model-based) NPP are shown along the y axis.

models, biases in NCP are consistent with biases in NPP, with the three MPI-ESM models overestimating NCP, and the two IPSL models underestimating it. This strong relationship motivates future CMIP experiments to include NCP as a direct output variable, enabling more robust atmospheric O_2 -based constraints on marine productivity.

Implications for modelling SO CO_2 uptake

Our NPP constraint provides a valuable benchmark for evaluating ocean carbon cycle simulations in CMIP6 models. Airborne CO_2 observations (Fig. 4a) and surface ocean pCO_2 observations are consistent in showing that the mid- to high-latitude SO is a strong CO_2 sink in the summer^{26,39–41}. However, CMIP6 models show significant spread and biases in simulating SO CO_2 uptake, with discrepancies in both uptake magnitude and seasonal phase (Fig. 4a). These biases are also evident in simulated surface ocean pCO_2 (Fig. 4b) that is used to calculate fluxes in the models. Decomposition of the pCO_2 seasonal cycle reveals that biases occur in both thermal and non-thermal components (Methods). For the non-thermal component (Fig. 4d), models underestimating NPP show insufficient September-to-February non-thermal pCO_2 drawdown. For the thermal component (Fig. 4c), most of the coupled simulations overestimate September-to-February thermal pCO_2 change due to overall warmer SST and excessively large SST seasonal amplitude (Extended Data Fig. 5a), driving excessive summer CO_2 outgassing.

We find that modelled summer air-sea CO_2 exchange correlates with summer NPP (Fig. 5a), indicating that production is a key factor determining summer CO_2 uptake. Most models underestimate this biological uptake, with summer net uptake further weakened in some models by excessive thermal CO_2 outgassing. The ten models with summer CO_2 uptake smaller than 0.1 PgC or showing outgassing

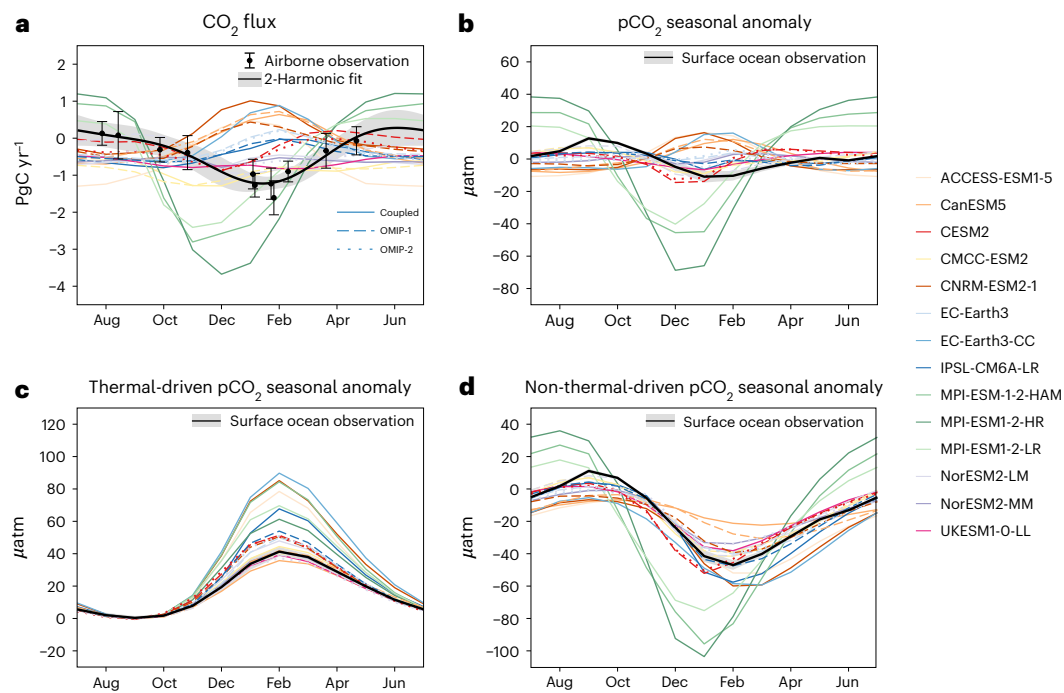


Fig. 4 | Observations and CMIP6 model simulations of seasonal air–sea CO_2 fluxes, surface ocean seasonal pCO_2 change, and its thermal and non-thermal components. **a**, Seasonal air–sea CO_2 fluxes, compared with airborne-based estimates (2009 to 2018 average) reported by Jin et al.²⁶ using the same airborne campaign data and method for calculating air–sea O_2 fluxes. Black points and error bars represent the mean and standard deviation of the flux ensemble ($n = 1,000$) for each campaign. The black line shows the mean of 2-harmonic fits across ensemble members, with grey shading indicating standard deviation. **b–d**, Model surface ocean seasonal pCO_2 change, and the thermal and non-thermal seasonal pCO_2 cycle, compared with observation-based seasonal

pCO_2 changes from a machine learning neural network product (SOM-FFN⁵⁰). The decomposition of seasonal pCO_2 changes into thermal and non-thermal components follows Takahashi et al.³¹, but uses seasonal SST anomaly referenced to September rather than the annual mean to focus on winter-to-summer changes (Supplementary Text 10). September was chosen as the reference because SST remains nearly constant from June to September before increasing rapidly (Extended Data Fig. 5a), indicating minimal thermal forcing and surface ocean equilibration through September. We use observed SST from EN4⁵². We use the same CMIP6 simulations as in Fig. 2. Observed pCO_2 seasonal cycles (**b–d**) are calculated as the average over 2005–2014.

(Fig. 5b) exhibit anticorrelated seasonal CO_2 uptake cycles compared with observation-based estimates, indicating too strong thermal control on air–sea CO_2 exchange. By contrast, the three MPI-ESM models consistently show excessive summer and annual NPP, leading to excessive biological CO_2 uptake that is partially offset by their overestimated summer thermal CO_2 outgassing (Fig. 5a). Among all models analysed, only NorESM2-LM (OMIP-1) simulates both summer NPP and CO_2 uptake in agreement with observation-based estimates.

Models with weak summer CO_2 uptake also show weak annual CO_2 uptake (Fig. 5b), highlighting the importance of understanding the competing thermal and non-thermal processes affecting the SO carbon budget. CMIP6 coupled simulations generally show lower ocean heat uptake than reanalysis⁴², but they exhibit exaggerated summer thermal CO_2 outgassing due to SST biases, indicating modelled biases in heat redistribution linked to circulation dynamics. These circulation biases could also affect the redistribution of macronutrients and iron, as well as stratification, potentially leading to errors in biological production related to biomass accumulation and nutrient and light utilization. Among models that output nitrogen, light and iron limitation factors, the model with NPP closest to our airborne-based estimate (CESM2) shows diatom growth colimited by light and iron during summer (Extended Data Fig. 5f–h) consistent with observations⁴³. However, models with low NPP (CanESM5 and IPSL-CM6A-LR) exhibit diatom growth colimitation by nitrogen and iron in summer. Circulation biases could arise from physical parameterizations or the atmospheric component of the coupled simulations. Additional model biases in CO_2 uptake might stem from gas-exchange velocity^{44,45} (Extended Data Fig. 5d), other biogeochemical and physical processes (for example, remineralization rates and depth, zooplankton grazing

and microbial pools) or insufficient spin-up periods that fail to bring ocean biogeochemical components to steady state⁴⁶.

Implications for twenty-first century climate projections

Understanding future SO productivity and CO_2 uptake changes is critical for climate projections. Our airborne-based constraints help to reduce uncertainty in future projections and provide a methodological framework for detecting long-term trends. We analyse CMIP6 projections under SSP3-7.0, a high-emissions scenario with global CO_2 reaching 867.2 ppm and radiative forcing reaching 7.0 W m^{-2} by 2100³². We find that models project consistently enhanced SO CO_2 uptake, with similar increases despite different present baseline uptake, indicating persistent systematic biases (Extended Data Fig. 6). By weighting the ten CMIP6 models according to their skill in reproducing observed NPP, annual CO_2 uptake and thermal-driven pCO_2 changes (Methods), we obtain a skill-constrained 2081–2100 annual SO CO_2 sink of $1.60 \pm 0.16 \text{ PgC yr}^{-1}$, compared with the unweighted ensemble mean of $1.54 \pm 0.34 \text{ PgC yr}^{-1}$, reducing uncertainty by 53%.

CMIP6 models, however, project diverse SNO_{bio} and NPP trends under SSP3-7.0, with most models showing increases, but two showing slight decreases (Extended Data Fig. 7). The annual NPP– SNO_{bio} coupling also varies substantially across models, with slopes ranging from 0.004 to $0.025 \text{ PgC Tmol}^{-1}$. These spreads indicate considerable uncertainty in model representations of SO biogeochemical responses to climate change. Despite this uncertainty, modelled SNO_{bio} trends correlate with NPP trends, motivating continued O_2 measurements on future airborne campaigns to constrain long-term SO NPP trends. Satellite-based NPP estimates diverge, showing trends of $0.8\% \text{ yr}^{-1}$ or

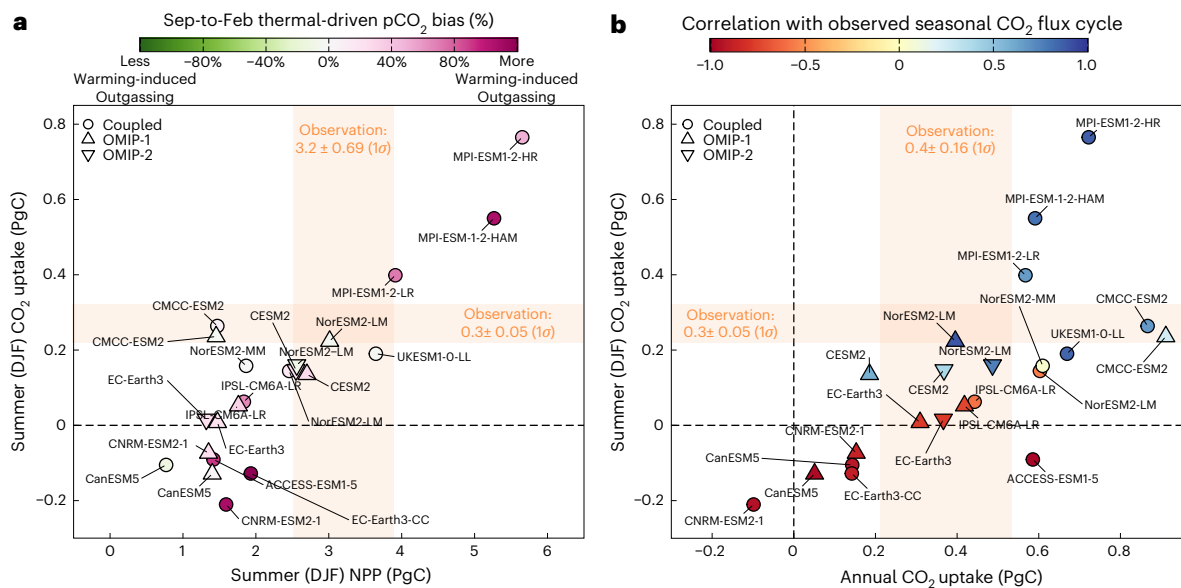


Fig. 5 | Analysis of thermal and non-thermal factors leading to biases in CMIP6 model simulations of SO CO₂ uptake. a, Examining factors contributing to biases in modelled summer-time CO₂ uptake. The x axis and y axis show summer-time (DJF) NPP and ocean CO₂ uptake, respectively, from CMIP6 (points) and observation-based estimates (orange shading, mean and standard deviation). The x-axis observation constraint (orange shading) is HEC-derived DJF NPP (Supplementary Fig. 5), while the y-axis constraint (orange shading) is airborne-based DJF CO₂ uptake from Fig. 4a. Marker colours indicate the percentage bias in the summer thermal-driven seasonal pCO₂, defined as the difference between February and September thermal-driven pCO₂ values compared with the estimate derived from surface ocean observations (Fig. 4c).

Note that biases in CO₂ fluxes may also arise from factors beyond NPP and thermal-driven pCO₂ changes. **b**, Comparing annual average CO₂ uptake (x axis, PgC) and summer-time (DJF) CO₂ uptake (y axis, PgC) for CMIP6 models. Orange shadings represent the mean and standard deviation of observation-based air–sea CO₂ fluxes in Fig. 4a. The marker colour reflects the temporal correlation between modelled and observed seasonal CO₂ flux cycles, with blue indicating in-phase correlation (positive) and red indicating 180° out-of-phase correlation (negative). Models with higher summer-time CO₂ uptake generally exhibit greater annual uptake. Coupled runs are shown as circles, OMIP-1 as upward triangles and OMIP-2 as downward triangles.

–0.5% yr^{–1} from 1997 to 2019 over the entire SO, depending on the productivity model used⁴⁷. Given the observed increase in high-latitude Chl-*a*⁴⁸, we use a hypothetical future NPP increase of 0.8% yr^{–1}, equivalent to an SNO_{bio} trend of 1.5 Tmol yr^{–1}. Achieving a 90% probability of detecting this trend at the 95% statistical confidence level requires at least two additional decadal sets of approximately ten global airborne surveys with spatial and temporal coverage similar to HIPPO, ORCAS and ATom (Methods). Earlier detection is achievable through denser spatial and temporal sampling and improved measurement calibration (Supplementary Text 11).

This study demonstrates that airborne O₂ observations can define SO O₂ fluxes, effectively constrain SO NPP and enable future trend detection, with direct implications for understanding SO CO₂ uptake and reducing uncertainty in model projections. Integrating this robust basin-scale constraint with BGC-Argo floats and ship-based measurements that capture fine-scale variability would improve our understanding of SO biogeochemical processes and generate more reliable future climate projections.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41561-026-01944-z>.

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Methods

Airborne and surface station $\delta(\text{O}_2/\text{N}_2)$ measurements

Atmospheric O_2 is measured as deviations in the O_2 to N_2 ratio relative to an arbitrary reference, expressed as $\delta(\text{O}_2/\text{N}_2)$. $\delta(\text{O}_2/\text{N}_2)$ is conventionally multiplied by 10^6 and expressed in units of per meg. The per meg unit can be converted to a ppm-equivalent unit by multiplying by X_{O_2} (0.2094), which is the reference dry-air mole fraction of O_2 used in the definition of the O_2/N_2 scale of the Scripps O_2 Program⁵³.

We use airborne $\delta(\text{O}_2/\text{N}_2)$ measurements made by the National Science Foundation's (NSF) National Center for Atmospheric Research (NCAR) Airborne Oxygen Instrument (AO2⁵⁴) collected from the NSF NCAR GV aircraft during the HIPPO Pole-to-Pole Observations (HIPPO³⁰) project from 2009 to 2011 and the O_2/N_2 Ratio and CO_2 Airborne Southern Ocean Study (ORCAS³¹) in January and February of 2016, as well as from the NASA DC-8 aircraft during the Atmospheric Tomography Mission (ATom³²) from 2016 to 2018. These airborne data typically span from the surface to above the tropopause, and for HIPPO and ATom, they provide full seasonal and global coverage (Fig. 1d and Supplementary Fig. 2). HIPPO consisted of five campaigns along a Pacific transect, ATom consisted of four campaigns covering both Pacific and Atlantic transects, and the 6-week ORCAS campaign provides dense sampling over the Drake Passage and ocean areas adjacent to the tip of South America and the Antarctic Peninsula (Supplementary Table 1). Here, we split ORCAS into three subcampaigns, each approximately 2 weeks long, but still refer to the three subcampaigns as campaigns.

The AO2 measurements were adjusted on a per-flight basis to match air samples collected during flight using the NSF NCAR/Scripps Medusa flask sampler, which collects 32 samples per flight for subsequent laboratory analysis at Scripps^{54,55}. These flask samples help to identify and correct time-dependent systematic biases in the continuous measurements. Before being used as a reference, the Medusa data underwent their own adjustment process to minimize diffusive fractionation effects inherent in flask sampling by applying an Ar/N_2 correction following

$$\delta\left(\frac{\text{O}_2}{\text{N}_2}\right)^* = \delta\left(\frac{\text{O}_2}{\text{N}_2}\right)_{\text{obs}} - \frac{1}{3.77} \times \delta\left(\frac{\text{Ar}}{\text{N}_2}\right)_{\text{obs}}, \quad (1)$$

where $\delta(\text{O}_2/\text{N}_2)_{\text{obs}}$ and $\delta(\text{Ar}/\text{N}_2)_{\text{obs}}$ represent Medusa flask measurements⁵⁴, and the coefficient 3.77 is the expected ratio between Ar/N_2 and O_2/N_2 variations resulting from thermal fractionation processes and is applied consistently across all campaigns⁵⁴. Therefore, our AO2 $\delta(\text{O}_2/\text{N}_2)$ measurements adjusted to match Medusa data are effectively $\delta(\text{O}_2/\text{N}_2)^*$, which includes a contribution from $\delta(\text{Ar}/\text{N}_2)$. $\delta(\text{O}_2/\text{N}_2)^*$ has reduced sensitivity to both thermal and pressure-gradient fractionation artefacts arising from airborne sampling and laboratory analysis. This adjustment does not contribute to the $\delta(\text{O}_2/\text{N}_2)$ long-term trend, but scales down the $\delta(\text{O}_2/\text{N}_2)$ seasonal cycle amplitude by about 5% (Extended Data Fig. 3). We correct for this impact on the net O_2 flux by directly adding the Ar/N_2 flux resolved by scaling ocean heat flux (see next section).

We quantify raw $\delta(\text{O}_2/\text{N}_2)^*$ measurement uncertainty by generating a large ensemble (1,000 members) of airborne data with four uncertainty components⁵⁴ (Supplementary Text 7), which are short-term instrument imprecision (± 2 per meg for each 10-s measurement), flight-to-flight reproducibility (± 1.4 per meg per flight), campaign-to-campaign reproducibility (± 3.1 per meg per campaign) and within-flight time-dependent biases (0.5 per meg h^{-1}).

We also utilize surface station $\delta(\text{O}_2/\text{N}_2)$ flask samples from seven sampling sites (Extended Data Fig. 1), primarily along the Pacific, within the Scripps O_2 Program surface flask network⁵³, to derive a deseasonalized cubic spline $\delta(\text{O}_2/\text{N}_2)$ trend (Supplementary Fig. 7). Flask uncertainty is quantified by generating a large ensemble of measurements

using an uncertainty model^{56,57} that considers a calibration scale span uncertainty of 1% and calibration cylinder corrosion uncertainty of ± 0.3 per meg yr^{-1} , leakage uncertainty of ± 0.2 per meg yr^{-1} , and desorption uncertainty of ± 0.1 per meg yr^{-1} . These potential systematic uncertainties are held constant across all stations for each ensemble member, and are referenced to 1 March 1989. In addition, we include random analysis uncertainty of ± 3.3 per meg and random thermal fractionation uncertainty of ± 2 per meg for each flask sample⁵⁷. The contribution of station flask samples uncertainty to the resulting flux uncertainty is negligibly small compared with the uncertainty in airborne measurements.

Deriving air–sea non-thermal O_2 flux from airborne $\delta(\text{O}_2/\text{N}_2)^*$ measurements

For each airborne campaign, we first calculate $\delta(\text{O}_2/\text{N}_2)^*$ flux ($F_{\text{O}_2}^{\text{raw}}$) south of -44°S (see spatial domain in Supplementary Figs. 8 and 9) from airborne $\delta(\text{O}_2/\text{N}_2)^*$ measurements using an isentrope-aligned box-inversion framework²⁶, centred on the average date of the campaign. We term it as $F_{\text{O}_2}^{\text{raw}}$ because it represents a combination of O_2 , N_2 and Ar fluxes (F_{O_2} , $F_{\text{N}_2}^{\text{ocn}}$ and $F_{\text{Ar}}^{\text{ocn}}$) that contribute to observed atmospheric $\delta(\text{O}_2/\text{N}_2)^*$ change. We then correct for the impact of $F_{\text{N}_2}^{\text{ocn}}$ and $F_{\text{Ar}}^{\text{ocn}}$ on $\delta(\text{O}_2/\text{N}_2)^*$, and correct for the impact of $F_{\text{N}_2}^{\text{ocn}}$ on the observations of $\delta(\text{Ar}/\text{N}_2)$ used to derive $\delta(\text{O}_2/\text{N}_2)^*$. The full correction equation is thus

$$F_{\text{O}_2} = F_{\text{O}_2}^{\text{raw}} + \frac{X_{\text{O}_2}}{X_{\text{N}_2}} F_{\text{N}_2}^{\text{ocn}} + \frac{1}{3.77} \left[\frac{X_{\text{O}_2}}{X_{\text{Ar}}} F_{\text{Ar}}^{\text{ocn}} - \frac{X_{\text{O}_2}}{X_{\text{N}_2}} F_{\text{N}_2}^{\text{ocn}} \right], \quad (2)$$

with full derivation in Supplementary Text 1. X_{N_2} (0.7808) and X_{Ar} (0.0094) are the dry mole fractions of N_2 and Ar in the atmosphere. We derive $F_{\text{N}_2}^{\text{ocn}}$ and $F_{\text{Ar}}^{\text{ocn}}$ (Extended Data Fig. 3) from ocean heat fluxes (see below).

The resolved F_{O_2} includes flux contributions at the air–sea interface ($F_{\text{O}_2}^{\text{ocn}}$), at the air–land interface ($F_{\text{O}_2}^{\text{land}}$) and from O_2 consumption due to fossil fuel burning ($F_{\text{O}_2}^{\text{ff}}$), following

$$F_{\text{O}_2} = F_{\text{O}_2}^{\text{ocn}} + F_{\text{O}_2}^{\text{land}} + F_{\text{O}_2}^{\text{ff}}. \quad (3)$$

South of -44°S and over the seasonal cycle, F_{O_2} is dominated by $F_{\text{O}_2}^{\text{ocn}}$, with a small contribution ($<1\%$) from combined $F_{\text{O}_2}^{\text{land}}$ and $F_{\text{O}_2}^{\text{ff}}$. We nevertheless correct these two impacts based on the posterior inverted terrestrial biosphere CO_2 flux from CarbonTracker 2022⁵⁸, and the prescribed prior fossil fuel CO_2 emissions used in CarbonTracker 2022, with both terms converted to O_2 flux (Supplementary Text 1). We show the seasonal cycle in Extended Data Fig. 3d.

$F_{\text{O}_2}^{\text{ocn}}$ is further separated into the thermal-driven ($F_{\text{O}_2}^{\text{ocn, therm}}$) and non-thermal-driven ($F_{\text{O}_2}^{\text{ocn, non-therm}}$) components, following

$$F_{\text{O}_2}^{\text{ocn}} = F_{\text{O}_2}^{\text{ocn, therm}} + F_{\text{O}_2}^{\text{ocn, non-therm}}. \quad (4)$$

with the thermal component resolved from scaling ocean heat flux (see below), and the non-thermal component as the residual.

We quantify the uncertainty of each individual non-thermal O_2 flux estimate and the seasonal flux cycle by generating an ensemble (1,000 iterations) of flux estimates (see details in Supplementary Text 9), allowing for uncertainty from these sources: (1) $\delta(\text{O}_2/\text{N}_2)^*$ measurements (Supplementary Text 7); (2) the correction for spatial bias due to insufficient airborne coverage (Supplementary Text 4); (3) the correction for temporal bias due to incomplete airborne seasonal coverage from 2009 to 2018 (Supplementary Text 5); (4) the diabatic mixing rates (Supplementary Text 6); and (5) the air–sea thermal-driven O_2 flux, N_2 flux and Ar/N_2 flux correction (see below). The 1σ uncertainty of the flux is therefore the standard deviation of the ensemble. We also derive the seasonal cycle of the non-thermal O_2 flux using a weighted

2-harmonic fit to the 12 flux estimates, with the 1σ uncertainty defined as the standard deviation of the fits to the ensembles (1,000 members per campaign) of flux estimates.

Isentrope-aligned box-inversion framework

We use the isentrope-aligned box model framework (Supplementary Fig. 1) developed by Jin et al.²⁶ to calculate climatological $F_{O_2}^{raw}$ south of $\sim 44^\circ$ S for each airborne campaign, centred at the campaign mean date. This approach builds upon a mass-indexed moist isentropic (θ_e) coordinate, M_{θ_e} , introduced by Jin et al.²⁸. The spatial domain of the flux resolved from this method is defined by surface M_{θ_e} contours (equivalent to θ_e), for which we use the 30 (10^{16} kg air mass) M_{θ_e} surface as the northern boundary here. A schematic of the zonal distribution of the 30 M_{θ_e} contour is shown in Supplementary Figs. 8 and 9, which changes over time, but has small seasonal meridional variability over the course of a year with a zonal average of approximately 44° S. The method allows $F_{O_2}^{raw}$ (F , Tmol yr⁻¹) to be computed from atmospheric O_2 mass balance of the overlying M_{θ_e} box but through conservation of tracer $\delta(O_2/N_2)^*$, based on the knowledge of atmospheric $\delta(O_2/N_2)^*$ inventory change ($\delta M/\delta t$, Tmol day⁻¹) and the diabatic transport (T , Tmol day⁻¹) of $\delta(O_2/N_2)^*$ across the M_{θ_e} box boundary, following

$$\frac{\partial M(t)}{\partial t} = F(t) + T(t). \quad (5)$$

Conceptually, the terms $\delta(O_2/N_2)^*$ inventory change and diabatic transport are defined as a scaled sum of O_2 , N_2 and Ar change according to their contribution to observed atmospheric $\delta(O_2/N_2)^*$ change. The inventory change term, at each airborne campaign mean date, is calculated as the time derivative of a daily time series. This time series combines a climatological seasonal cycle derived from a 2-harmonic fit to detrended airborne $\delta(O_2/N_2)^*$ measurements, and a cubic spline trend derived from surface station $\delta(O_2/N_2)$ measurements of the Scripps O_2 Program (Supplementary Fig. 7). The concentration values are converted to mole fraction units assuming constant N_2 and then to O_2 mass units by multiplying by the total dry air mass of the M_{θ_e} box (that is, 30×10^{16} kg in this study). Details are presented in Supplementary Text 2.

The cross- M_{θ_e} transport term (T) is considered as a one-dimensional diffusive system (poleward positive) that is parameterized according to

$$T(t) = D(t) \times \frac{\Delta\chi(t)}{\Delta M_{\theta_e}} \times \kappa, \quad (6)$$

where $\Delta\chi(t)$ (per meg) is the $\delta(O_2/N_2)^*$ gradient across the M_{θ_e} boundary (30 surface). The gradient is calculated as $\delta(O_2/N_2)^*$ of the 30–45 (10^{16} kg) M_{θ_e} band minus that of the 15–30 (10^{16} kg) M_{θ_e} band (Supplementary Text 3). ΔM_{θ_e} is therefore defined as the distance in M_{θ_e} coordinates between the two bands used to calculate the gradient (that is, 15×10^{16} kg in this study). κ is a constant to convert the unit from per meg to Tmol. For both inventory change and gradient, we correct for small biases induced by sparse spatial coverage of the airborne observations (Supplementary Text 4 and Supplementary Fig. 10).

D is the diabatic mixing rate ($(10^{16} \text{ kg})^2 \text{ day}^{-1}$) that represents the mixing timescale across the 30 (10^{16} kg) M_{θ_e} boundary. Here, we diagnose diabatic mixing rates (monthly climatology) from the moist static energy budget of four reanalyses (Supplementary Text 6 and Supplementary Fig. 12), which are ERA-5⁵⁹, JRA-3Q⁶⁰, JRA-55⁶¹ and MERRA-2⁶², consistent with the rates used by Jin et al.²⁷. This method quantifies large-scale mixing without using atmospheric transport models, which were shown to have large error, typically overestimating mixing rates over the mid-latitudes of the Southern Hemisphere²⁶. We fit a 2-harmonic curve to monthly mixing rates and sample the rates on each airborne campaign mean date as the input to the model. We report the resolved flux and each component for each airborne campaign in Supplementary Tables 3 and 4.

Air–sea N_2 , Ar and thermal-driven O_2 flux

$F_{N_2}^{ocn}$, F_{Ar}^{ocn} and $F_{O_2}^{ocn}$ (mol m⁻² s⁻¹, positive into the atmosphere) are calculated following

$$F_{tracer}^{ocn}(t) = \left(\frac{dS}{dT}(t) \right)_{tracer} \times \frac{Q(t)}{C_p}, \quad (7)$$

where dS/dT (mol kg⁻¹ C⁻¹) is the temperature derivative of solubility of Ar, N_2 or O_2 (ref. 63). The calculation of dS/dT requires SST and sea surface salinity, for which we use monthly EN4 data⁵², interpolated to each reanalysis grid. Q is the ocean heat flux (W m⁻², positive into the ocean), for which we alternatively use four reanalyses (that is, ERA-5, JRA-3Q, JRA-55 and MERRA-2). C_p represents the specific heat capacity of seawater, which is assumed to be $3,993 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$.

The resolved flux is further corrected to reduce the seasonal amplitude by a factor of 1.3 and shift the phase later by 0.5 months, as proposed by an earlier study⁶⁴, to account for biases from gas exchange rates in the box model used by Keeling and Shertz³⁶. This correction follows

$$F(t) = \frac{1}{1.3} \times F_{tracer}^{ocn}(t - 0.5). \quad (8)$$

We show the monthly average fluxes of these three components in Extended Data Fig. 3. The uncertainty of these three fluxes incorporates the spread of ocean heat flux across four reanalyses and uncertainty in the 1/1.3 scaling factor. Because this scaling factor is derived from a single model, we assume a $\pm 15\%$ uncertainty.

CMIP6

Our analysis of CMIP6 models uses coupled historical simulations and OMIP (OMIP-1 and OMIP-2) simulations. Coupled historical runs simulate the Earth's climate from 1850 to 2014 using coupled Earth system models driven by prescribed atmospheric concentrations of greenhouse gases (for example, CO_2), alongside other forcings such as solar irradiance, stratospheric aerosol, SSTs, sea-ice concentrations and land-use change⁷. OMIP-1 and OMIP-2 are ocean-only hindcast simulations that evaluate ocean physics and biogeochemistry, forced by prescribed surface CO_2 concentrations and atmospheric state variables^{65,66}. OMIP-1 is forced by atmospheric state from 1948 to 2009, following the protocol of the interannual Coordinated Ocean-ice Reference Experiments (CORE-II⁶⁷). OMIP-2 is forced by atmospheric state from JRA-55 reanalysis from 1958 to 2018. Both simulations are repeated over multiple cycles of the time window to reach a quasi-steady state. Supplementary Table 2 lists the models and experiments analysed in this study. Models were selected on the basis of the availability of air–sea O_2 flux (fgo2) and depth-integrated primary production (intpp) data. CMIP6 NCP estimates are based on the variable 'fbddtdic', which is defined as rate of change of DIC due to biological activity. We also analyse output from SSP3-7.0 simulations, which project climate and biogeochemical changes from 2015 to 2100 using fully coupled Earth system models, driven by prescribed greenhouse gas emissions, land-use trajectories and air pollutant emissions consistent with a high-emissions pathway characterized by regional rivalry, slow technological development and limited climate policy⁶⁸. SSP3-7.0 represents one of the more pessimistic scenarios in CMIP6, with CO_2 emissions continuing to rise throughout the twenty-first century.

A hierarchical statistical framework for emergent constraints

The HEC framework³³ leverages the relationship between the modelled annual depth-integrated NPP (y^{model}) and SNO_{bio} (x^{model}), constrained by airborne observation of SNO_{bio} (x^{obs}). When Gaussian assumptions are made, the conditional mean and variance of NPP given observed SNO_{bio} is

$$E(y^{\text{model}}|x^{\text{obs}}) = \mu_{y^{\text{model}}} + K_{y^{\text{model}}, x^{\text{model}}} \times K_{x^{\text{model}}, x^{\text{obs}}}^{-1} \times (x^{\text{obs}} - \mu_{x^{\text{obs}}}) \quad (9)$$

and

$$\text{var}(y^{\text{model}}|x^{\text{obs}}) = \sigma_{y^{\text{model}}}^2 - K_{y^{\text{model}}, x^{\text{model}}}^2 \times K_{x^{\text{model}}, x^{\text{obs}}} \times \sigma_{x^{\text{model}}}^2, \quad (10)$$

where the gain of the conditional expectation (K) is expressed as

$$K_{x^{\text{model}}, x^{\text{obs}}} = \frac{\sigma_{x^{\text{model}}}^2}{\sigma_{x^{\text{model}}}^2 + \sigma_{x^{\text{obs}}}^2} \quad (11)$$

and

$$K_{y^{\text{model}}, x^{\text{model}}} = \frac{\rho \sigma_{y^{\text{model}}}}{\sigma_{x^{\text{model}}}}. \quad (12)$$

In equation (12), ρ is the correlation between NPP and SNO_{bio} across CMIP6 models.

The constrained NPP variance reduction factor is calculated as

$$\frac{\sigma_{y^{\text{model}}|x^{\text{obs}}}^2}{\sigma_{y^{\text{model}}}^2} = 1 - \frac{\rho^2}{1 + \left(\frac{\sigma_{x^{\text{model}}}^2}{\sigma_{x^{\text{obs}}}^2}\right)^{-1}}. \quad (13)$$

A full derivation of these equations can be found in the work of Bowman et al.³³.

We construct the relationship between SNO_{bio} and annual depth-integrated NPP (Fig. 3), summer (December–January–February, DJF) depth-integrated NPP (Supplementary Fig. 5) and growing-season NCP (Extended Data Fig. 4) in the SO (south of ~44° S), using available CMIP6 models and experiments. We use airborne observation of SNO_{bio} to provide strong constraints on NPP or NCP. This approach allows us to incorporate both observational constraints and model-based relationships within a consistent Bayesian framework.

Constraining future SO CO₂ uptake

To assess the implications of CMIP6 model biases for projecting future CO₂ uptake, we constrain their end-of-century (2081–2100 average) annual SO CO₂ sink estimates using model skill in simulating present-day conditions in historical coupled runs. We evaluate ten CMIP6 models that provide both SSP3-7.0 and historical coupled runs (Supplementary Table 2) based on their ability to reproduce three metrics, which are annual depth-integrated NPP derived from airborne O₂ measurements in this study ($k=1$), annual CO₂ uptake from the similar analysis of airborne CO₂ measurements²⁶ ($k=2$) and September-to-February thermal-driven seasonal pCO₂ change estimated from observed SST⁵² and pCO₂⁵⁰ ($k=3$), during the time window of 2009 to 2018 (Fig. 5). For each model (i), we calculate a normalized and squared bias (s_i), following

$$s_i = \sum_{k=1}^3 \left[\frac{m_{k,i}^{\text{model}} - m_k^{\text{obs}}}{\sigma_{m_k^{\text{obs}}}} \right]^2, \quad (14)$$

where i is the index of the model, m_k^{model} and m_k^{obs} are model and observed values for the k th metric, and σ represents the observational uncertainty of each metric.

To calculate an observation-constrained model mean, we weight each of the CMIP6 models according to

$$w_i = \frac{s_i}{\sum_{i=1}^{10} s_i}. \quad (15)$$

We report the normalized and squared bias of each metric and the resulting weights of each model in Supplementary Table 2.

The weighted average ocean CO₂ uptake (F^{CO_2}) is therefore calculated as $\sum_{i=1}^{10} w_i \times F_i^{\text{CO}_2}$, with 1σ uncertainty calculated as $\sqrt{\sum_{i=1}^{10} [w_i (F_i^{\text{CO}_2} - F^{\text{CO}_2})^2]}$.

Detection of trends

We adopt the method of Weatherhead et al.⁶⁹ to estimate the minimum required time for the expected linear trend in SNO_{bio} to emerge at a statistically significant level. This method is designed to estimate the record length required to achieve a 90% probability of detecting a real trend that is indicated at a 95% confidence level, following

$$N^{\frac{3}{2}} = \frac{\sigma \times 3.3}{|w_0| \times \Delta t} \times \sqrt{\frac{1+\alpha}{1-\alpha}}, \quad (16)$$

where N is the required number of measurements, 3.3 is the confidence factor for 90% certainty, σ is the uncertainty in SNO_{bio} estimates, w_0 is the expected linear trend, Δt is the measurement interval and α is the lag-1 autocorrelation coefficient, which is assumed to be 0 here. With airborne campaigns that have similar spatial and temporal coverage as HIPPO, ORCAS and ATom, we can obtain a decadal-averaged SNO_{bio} with uncertainty (σ) of 25.2 Tmol yr⁻¹, and Δt of 10 years. Given a trend of 1.5 Tmol yr⁻¹, derived from a hypothetical future NPP increase of 0.8% yr⁻¹, equation (16) yields $N = 3.14$. Because N represents the total required measurements in the time series, this result indicates that approximately three evenly spaced decadal averages are required. Given that the initial measurement is already available, this corresponds to approximately two additional decades of measurements that have spatial and temporal coverage similar to that of HIPPO, ORCAS and ATom, to achieve a 90% probability of detecting the 1.5 Tmol yr⁻¹ trend at the 95% confidence level. The potential for earlier trend detection is discussed in Supplementary Text 11.

Data availability

All HIPPO 10-s merged data are available at <https://data.eol.ucar.edu/dataset/112.123> (ref. 70). We use updated HIPPO AO2 data from <https://data.eol.ucar.edu/dataset/112.005> (ref. 71), <https://data.eol.ucar.edu/dataset/117.082> (ref. 72), <https://data.eol.ucar.edu/dataset/121.009> (ref. 73), <https://data.eol.ucar.edu/dataset/248.010> (ref. 74) and <https://data.eol.ucar.edu/dataset/249.010> (ref. 75). All ORCAS 10-s merge data are available at <https://data.eol.ucar.edu/dataset/490.024> (ref. 76). Here, we use updated ORCAS AO2 data from <https://data.eol.ucar.edu/dataset/490.015> (ref. 77). All ATom 10-s merge data are available at https://daac.ornl.gov/cgi-bin/dsviewer.pl?ds_id=1925 (ref. 78), including the version of AO2 data used here. Ground station O₂/N₂ measurements from the Scripps O₂ Program are available at <https://doi.org/10.6075/JOWS8RJR> (ref. 53). CMIP6 data are available through the Earth System Grid Federation (ESGF; <https://esgf-node.llnl.gov/projects/esgf-llnl/>). Surface ocean pCO₂ data are downloaded at https://www.ncei.noaa.gov/access/ocean-carbon-acidification-data-system/oceans/SPCO2_1982_present_ETH_SOM_FFN.html (ref. 50). SST and sea surface salinity data from EN4 are downloaded at <https://www.metoffice.gov.uk/hadobs/en4/>. Jena APO inversion data are available at <https://www.bgc-jena.mpg.de/CarboScope/?ID=apo>. Simulations of APO components from APO-MIP1 are available at <https://gdex.ucar.edu/datasets/d010018/> (ref. 79). The data needed to reproduce Figs. 1–5 and Extended Data Figs. 1–7 are available via Zenodo at <https://doi.org/10.5281/zenodo.17969932> (ref. 80).

Code availability

The code needed to reproduce Figs. 1–5 and Extended Data Figs. 1–7 is available via Zenodo at <https://doi.org/10.5281/zenodo.17969932> (ref. 80).

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Author contributions

Y.J. conceived of the project, conducted the analysis, generated the figures and wrote the paper. B.S. and M.L. contributed to the organization of the paper. B.S., E.M., and R.K. provided O₂ measurements. M.M., N.L., and C.N. contributed to the analysis of model simulations. All authors contributed to reviewing and editing the text.

Competing interests

The authors declare no competing interests.

Additional information

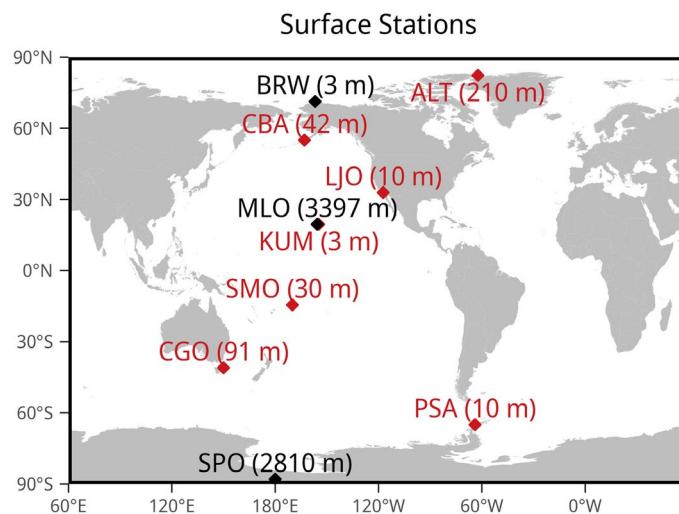
Extended data is available for this paper at <https://doi.org/10.1038/s41561-026-01944-z>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41561-026-01944-z>.

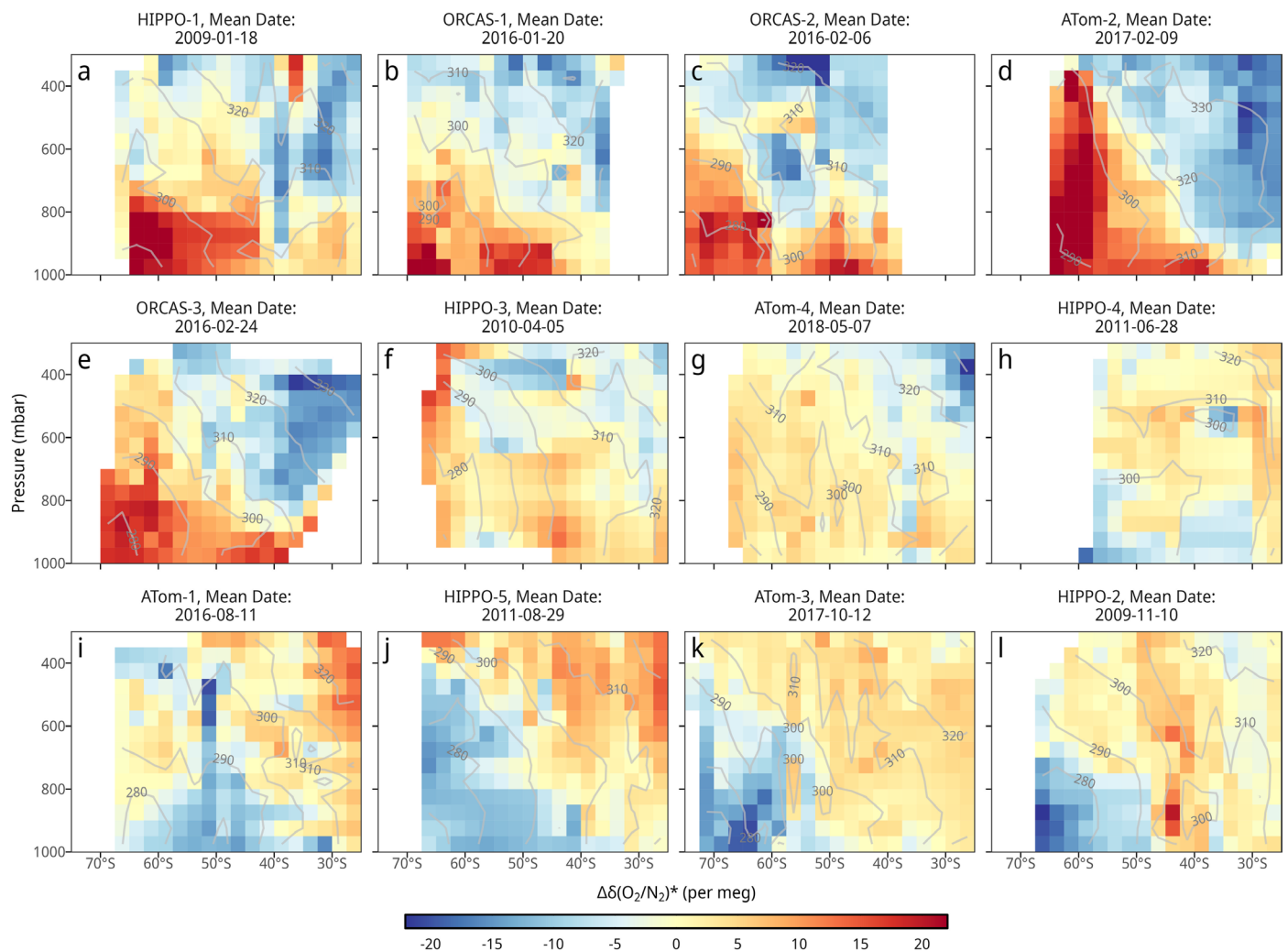
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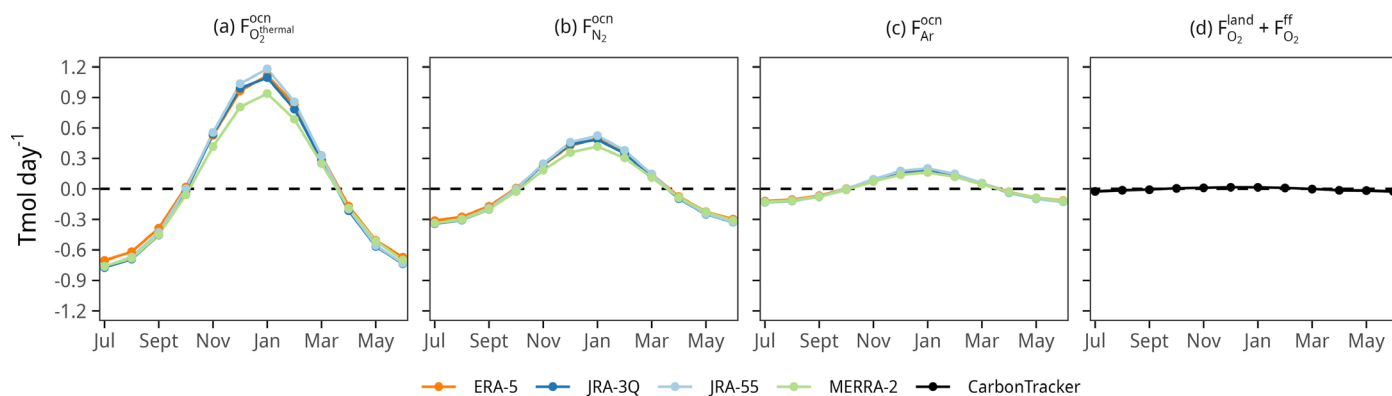


Extended Data Fig. 1 | Location of station measurements from the Scripps O₂ program stations with station codes and inlet elevations in meters above sea level. We only use data from stations marked in red to derive the cubic spline trend. Figure adapted from ref. [27](#) under a Creative Commons licence [CC BY 4.0](#).



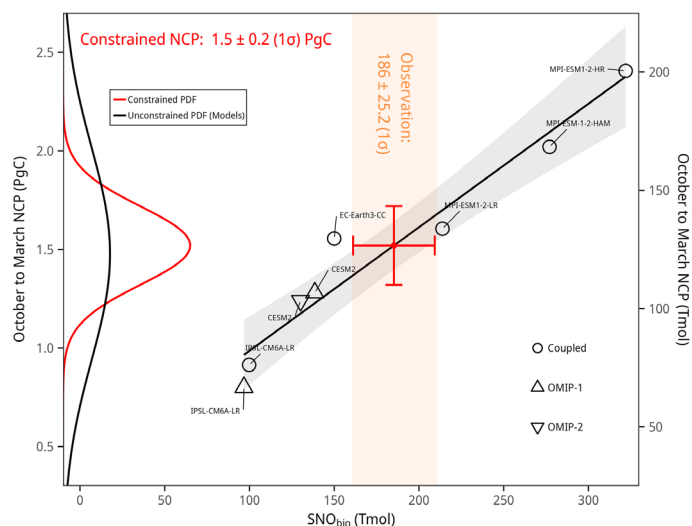
Extended Data Fig. 2 | Cross sections of airborne $\delta(\text{O}_2/\text{N}_2)^*$ observations of 12 campaigns. **a.** HIPPO-1 (mean date: 2009-01-18). **b.** ORCAS-1 (2016-01-20). **c.** ORCAS-2 (2016-02-06). **d.** ATom-2 (2017-02-09). **e.** ORCAS-3 (2016-02-24). **f.** HIPPO-3 (2010-04-05). **g.** ATom-4 (2018-05-07). **h.** HIPPO-4 (2011-06-28). **i.** ATom-1 (2016-08-11). **j.** HIPPO-5 (2011-08-29). **k.** ATom-3 (2017-10-12). **l.** HIPPO-2 (2009-11-10). Colors show the observed $\delta(\text{O}_2/\text{N}_2)^*$ averaged into 2.5° latitude

and 50-mbar pressure boxes with campaign averages (75–25°S, 1000–300 mbar) removed. Panels are ordered by the mean date of the year of each campaign. For boxes with fewer than 7 observations, we extrapolated using the Akima function (ref. 49) only for visualization purposes in the figure. Gray contour lines show the observed θ_e . Numbers of 10-s measurements within each latitude-pressure box are shown in Supplementary Fig. S1.



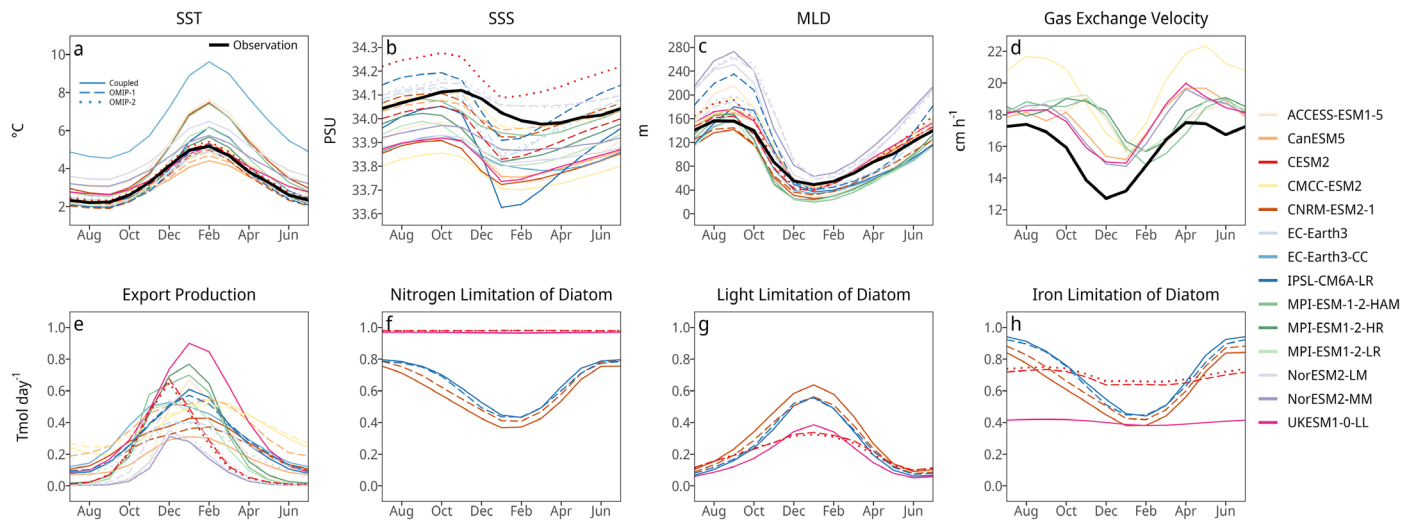
Extended Data Fig. 3 | Flux components used to correct the observed surface seasonal $\delta(O_2/N_2)$ flux cycle to derive the seasonal non-thermal air-sea O_2 flux cycle. (a–c) Monthly average air-sea thermal-driven O_2 flux, air-sea N_2 flux, and air-sea Ar flux calculated using ocean heat flux from four reanalyses, averaged from 2009 to 2018. (d) Sum of monthly average (2009 to 2018) air-land O_2 flux

and O_2 fossil fuel burning consumption calculated using monthly average posterior land CO_2 flux and prior fossil fuel CO_2 emission from CarbonTracker 2022, converted to O_2 flux. By comparison, the net $F_{O_2}^{ocn}$ has a seasonal cycle amplitude of 5.0 Tmol day⁻¹. The correction is presented in Supplementary Eq. S3.



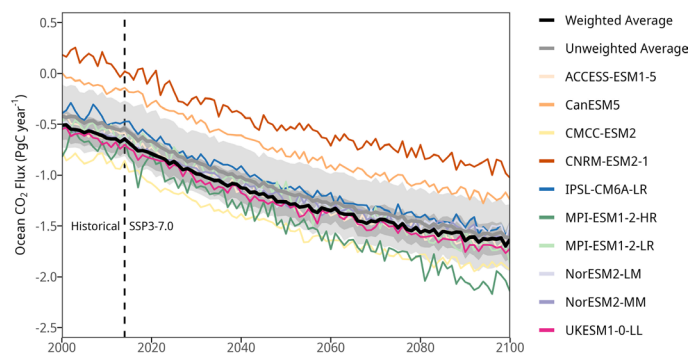
Extended Data Fig. 4 | Emergent constraint on growing season (October to March) Southern Ocean (SO) net community production (NCP) using the seasonal net outgassing of the non-thermal O_2 flux cycle (SNO_{bio}). We use the variable 'fbddtdic' as a proxy for NCP, defined as the rate of change of DIC due to biological activity, which is available in only 8 of the 22 models and experiments analyzed here. The emergent constraint method is identical to that reported in Fig. 3. The red point with error bars denotes the mean and standard deviation of constrained NCP ($1.5 \pm 0.20 \text{ PgC yr}^{-1}$) and SNO_{bio} derived using

HEC (Methods). We focus specifically on growing season (October to March) NCP because SNO_{bio} measures growing season net O_2 outgassing that directly relates to biological activities. During winter, negative NCP from respiration and limited photosynthesis does not correlate with SNO_{bio} . We note that previous studies suggest that the model variable 'epc100' (Extended Data Fig. 5e), which represents the export production at 100-m, can serve as a proxy for NCP, but significant uncertainty arises from temporal variations in organic carbon storage across models (refs. 14,20). Therefore, we do not use 'epc100' here.



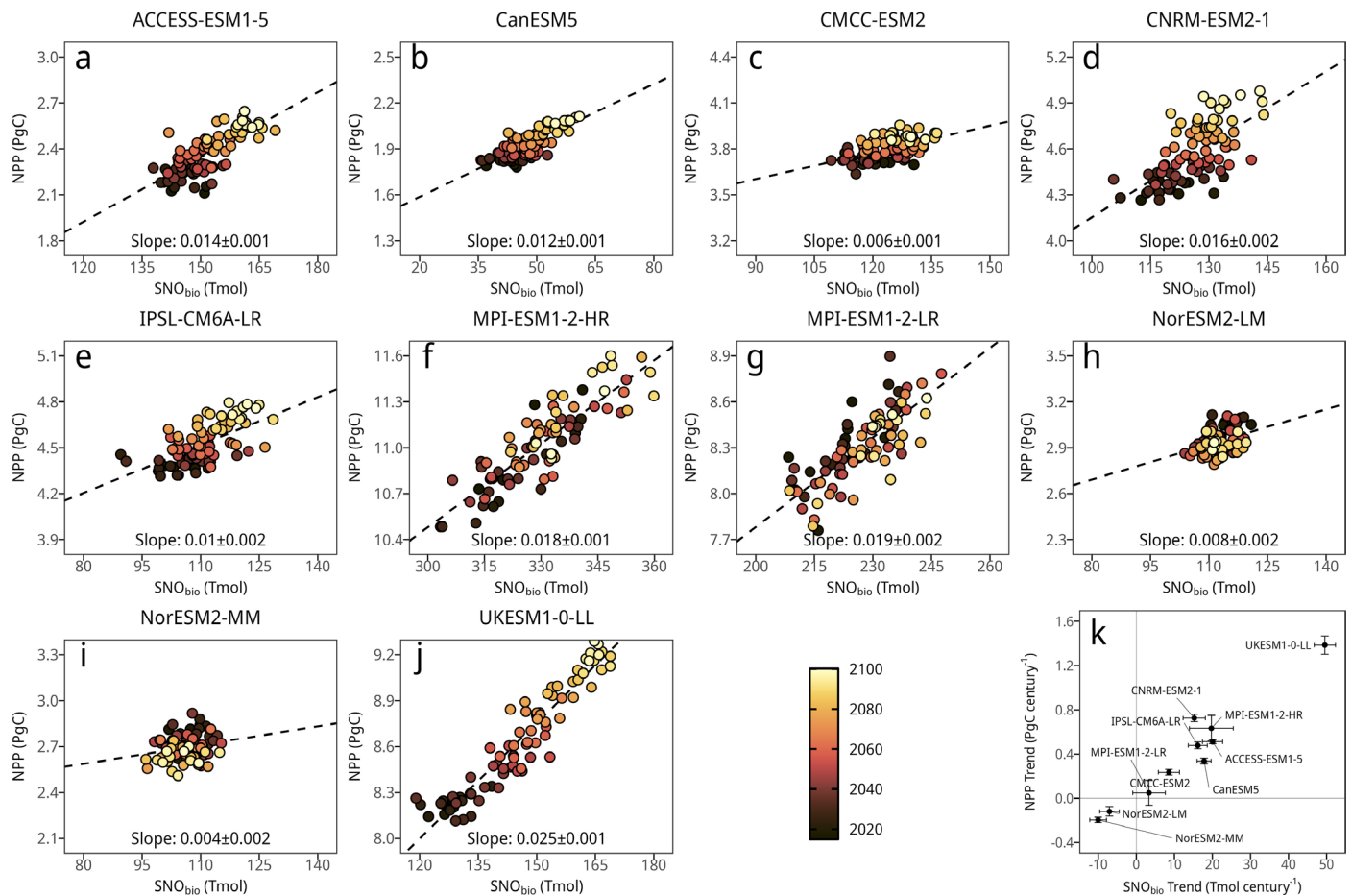
Extended Data Fig. 5 | CMIP6 model simulations of selected physical and biogeochemical variables. We show observations and CMIP6 simulations or diagnoses of (a) SST, (b) SSS, (c) MLD, (d) gas-exchange velocity. We also show CMIP6 simulations of (e) 100-m export production, and (f–h) nitrogen, light, and iron limitation of diatom integrated over 100-m from 90°S to ~44°S (defined by $0.30 \times 10^{16} \text{ M}_{\text{Diatom}}$). For (a) to (d), we compare model simulations with observations (black lines). We use observations of SST and SSS from EN4 (ref. 52), density MLD from de Boyer Montégut et al. (ref. 38) (climatological average from 1963 to 2008), gas exchange velocity calculated from CCMP (ref. 44) wind fields and the method of Wanninkhof (ref. 45). CMIP6 model simulations of MLD are calculated

using the same density criteria, defined as the depth at which the potential density exceeds the potential density at 10 meters depth by 0.03 kg m^{-3} . The gas exchange velocity for models is calculated using simulated surface wind speed with the same method as for observation. Note that OMIP models are driven by prescribed wind speed but these wind speeds are not available as model output. We use CMIP6 simulations from coupled runs (solid lines), OMIP-1 (dashed lines), and OMIP-2 (dotted lines), with each model represented by a distinct color (see Supplementary Table S2 for model details). Model seasonal cycles are monthly averages from 2005–2014 (coupled runs and OMIP-2) or 2005–2009 (OMIP-1). Observations are calculated as 2005 to 2014 averages.



Extended Data Fig. 6 | Skill-based model weighting reduces uncertainty in end-of-century CO₂ uptake projections. Annual mean SO CO₂ flux (negative values indicate uptake) from 10 CMIP6 models during historical (2000–2014) and SSP3-7.0 (2015–2100) periods. The thick black line shows the skill-weighted ensemble mean, where weights are based on model skill in reproducing observed NPP, annual CO₂ uptake, and thermal-driven pCO₂ changes during 2005–2014 (shown in Fig. 5), with the narrow dark gray shading indicating weighted 1σ

uncertainty (Methods). The thick gray line shows the unweighted ensemble mean annual CO₂ uptake, with the wide light gray shading indicating unweighted 1σ standard deviation across all models shown here. Models maintain near-constant offsets throughout the century, indicating that present-day biases persist systematically into future projections. The skill-weighted 2081–2100 mean is $1.60 \pm 0.16 \text{ PgC yr}^{-1}$, compared to an unweighted ensemble mean of $1.54 \pm 0.34 \text{ PgC yr}^{-1}$, reducing uncertainty by 53%.



Extended Data Fig. 7 | CMIP6 models show different trends in net primary production (NPP) and the seasonal net outgassing of the non-thermal O₂ flux cycle (SNO_{bio}) over the Southern Ocean (SO) under the SSP3-7.0 scenario. (a–j) Within-model relationships between annual SNO_{bio} and NPP for ten CMIP6 models over 2015-2100. Points are colored by year (darker = earlier, lighter = later), showing the temporal evolution of both variables. Dashed lines show fitted linear relationships using the ordinary least square fit method, with slopes (PgC Tmol⁻¹) and standard errors reported for each model. **(k)** Comparison of SNO_{bio} trends (x-axis, Tmol century⁻¹) versus NPP trends (y-axis, PgC century⁻¹)

across models. The black points and error bars represent the mean and standard deviation of trends calculated using generalized least squares regression with AR(1) correlation structure to account for temporal autocorrelation. Models show diverse future trajectories, with most models exhibiting positive SNO_{bio} and NPP trends and only two models showing negative or near-zero trends. Large spreads in projected trends and SNO_{bio}-NPP coupling across CMIP6 models highlight substantial uncertainty in Earth system model representations of SO biogeochemical responses to climate change.